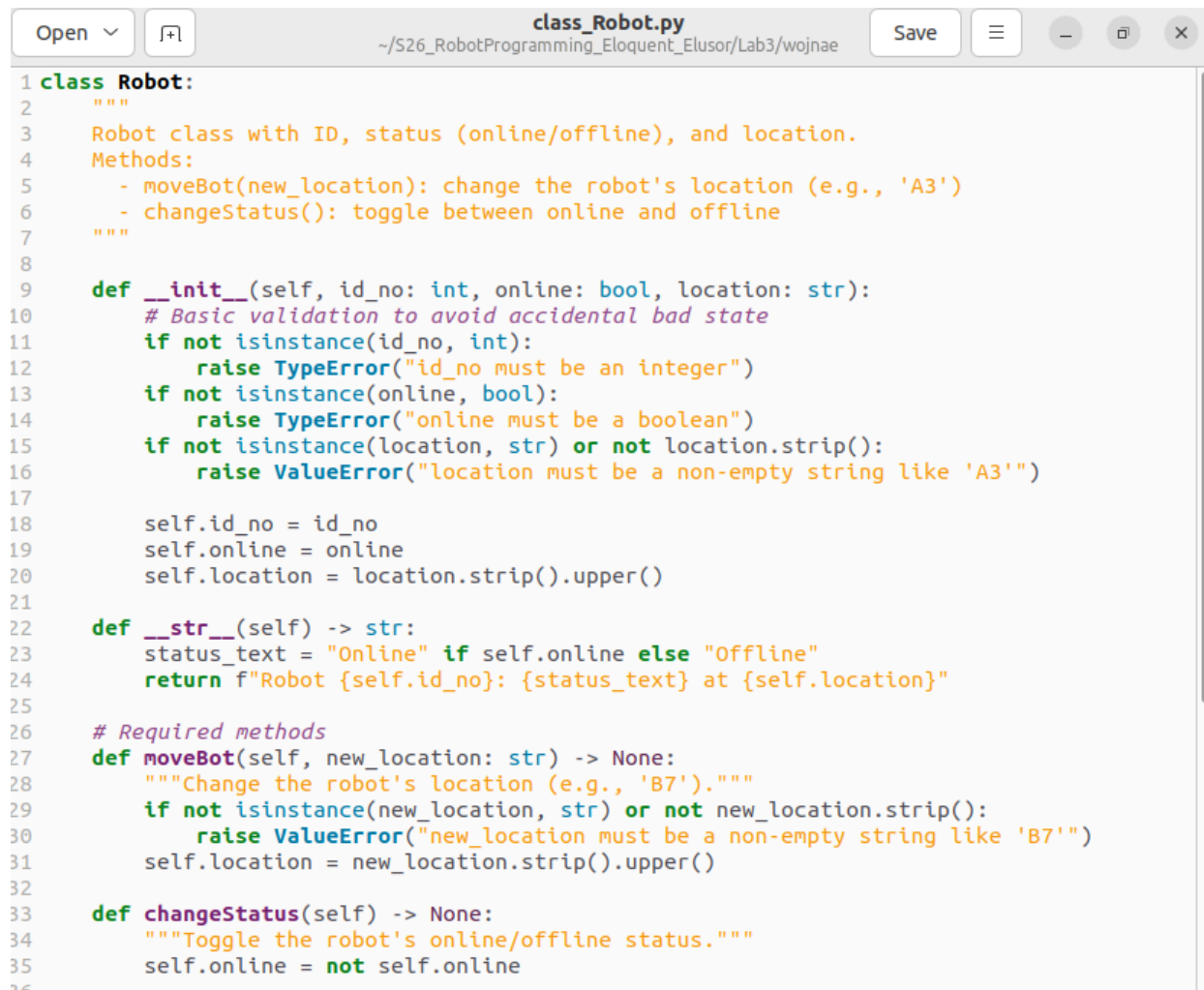


Lab 3 Individual Prelab

Robot Class:

Screenshot of Attributes and Methods:

A screenshot of a code editor window titled 'class_Robot.py'. The editor shows the implementation of a Python class named 'Robot'. The class has three attributes: 'id_no', 'online', and 'location'. It has five methods: '__init__', '__str__', 'moveBot', and 'changeStatus'. The code is written in Python 3 syntax with color-coding for keywords, strings, and comments. The window has a menu bar with 'Open', 'Save', and a hamburger menu icon. The file path is '~/.S26_RobotProgramming_Eloquent_Elutor/Lab3/wojnae'.

```
1 class Robot:
2     """
3     Robot class with ID, status (online/offline), and location.
4     Methods:
5     - moveBot(new_location): change the robot's location (e.g., 'A3')
6     - changeStatus(): toggle between online and offline
7     """
8
9     def __init__(self, id_no: int, online: bool, location: str):
10         # Basic validation to avoid accidental bad state
11         if not isinstance(id_no, int):
12             raise TypeError("id_no must be an integer")
13         if not isinstance(online, bool):
14             raise TypeError("online must be a boolean")
15         if not isinstance(location, str) or not location.strip():
16             raise ValueError("location must be a non-empty string like 'A3'")
17
18         self.id_no = id_no
19         self.online = online
20         self.location = location.strip().upper()
21
22     def __str__(self) -> str:
23         status_text = "Online" if self.online else "Offline"
24         return f"Robot {self.id_no}: {status_text} at {self.location}"
25
26     # Required methods
27     def moveBot(self, new_location: str) -> None:
28         """Change the robot's location (e.g., 'B7')."""
29         if not isinstance(new_location, str) or not new_location.strip():
30             raise ValueError("new_location must be a non-empty string like 'B7'")
31         self.location = new_location.strip().upper()
32
33     def changeStatus(self) -> None:
34         """Toggle the robot's online/offline status."""
35         self.online = not self.online
```

Screenshot of Script to Verify the Class Works:

```
37 if __name__ == "__main__":
38     # Create two robots
39     r1 = Robot(id_no=101, online=True, location="A3")
40     r2 = Robot(id_no=202, online=False, location="C5")
41
42     # Print initial state (__str__ used here)
43     print("Initial states:")
44     print(r1) # Robot 101: Online at A3
45     print(r2) # Robot 202: Offline at C5
46
47     # Move robot 1 and toggle robot 2
48     r1.moveBot("B7")
49     r2.changeStatus()
50
51     print("\nAfter updates:")
52     print(r1) # Robot 101: Online at B7
53     print(r2) # Robot 202: Online at C5
54
55     # Simple checks (won't print anything if all good)
56     assert r1.location == "B7"
57     assert r2.online is True
58
59     # Edge case demo (uncomment to see validation in action)
60     # Robot("BAD", True, "A1") # TypeError
61     # r1.moveBot("") # ValueError
```

Python 2 ▾ Tab Width: 8 ▾ Ln 30, Col 62 ▾ INS

Screenshot of Output:

```
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae$
python3 class_Robot.py
Initial states:
Robot 101: Online at A3
Robot 202: Offline at C5

After updates:
Robot 101: Online at B7
Robot 202: Online at C5
```